Adaptive Control Assignment

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Q1

The dynamic equation of this model is given by:

$$\dot{x_p} = -x_p + u \tag{1}$$

The second model for the dynamic system with the proposed value $a_m = -2$ is:

$$\dot{\hat{x_p}} = a_m \hat{x_p} + (\hat{a_p} - a_m) x_p + \hat{k_p} u \tag{2}$$

The estimation error is defined as:

$$e = \hat{x_p} - x_p \tag{3}$$

Adaptation laws:

$$\dot{\hat{a_p}} = -\gamma_1 e x_p \quad , \quad \dot{\hat{k_p}} = -\gamma_2 e u \tag{4}$$

To determine the optimal values of adaptation gains γ_1 and γ_2, we increase them in a way that minimizes the settling time of the system.

Also, since we have two parameters to estimate, a single sinusoidal input will be sufficient.

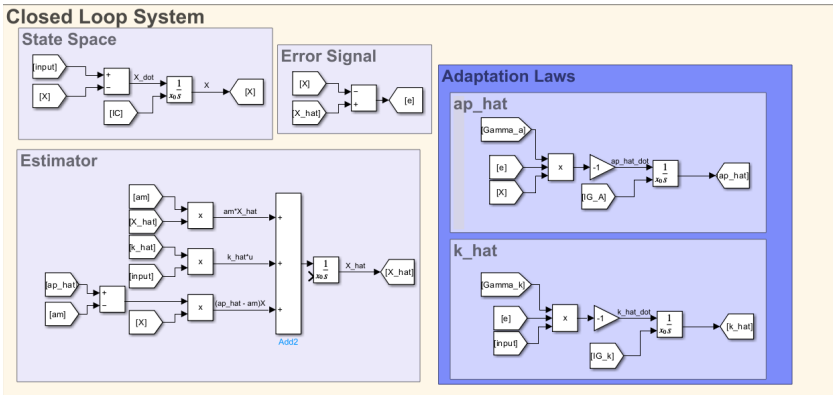


Figure 1: Overall closed-loop structure for Question 1 in Simulink

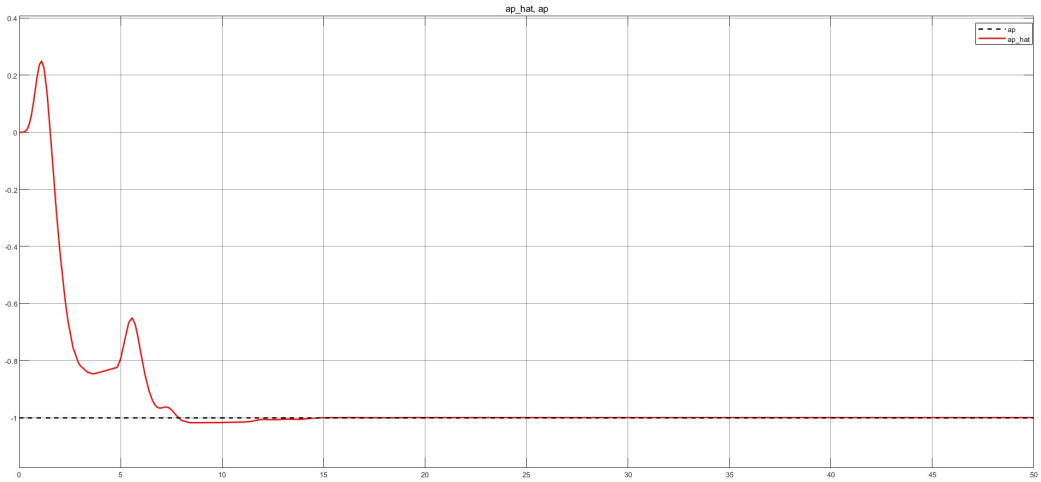


Figure 2: Parameter estimation $\hat{a_p}$

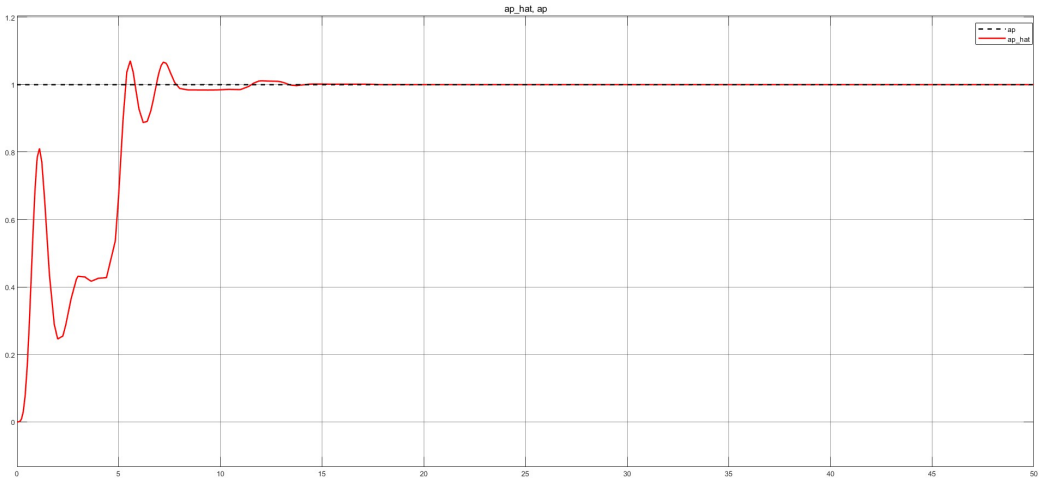


Figure 3: Parameter estimation $\hat{k_p}$

Q2

The dynamic equation of the system in vector form is:

$$\dot{X}_p = \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix} X_p + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \tag{5}$$

The proposed second model for the dynamic system:

$$A_m = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

The estimation equations are:

$$\dot{\hat{X}}_p = A_m \hat{X}_p + (\hat{A}_p - A_m) X_p + \hat{B}_p u \tag{6}$$

The estimation error is:

$$e = \hat{X}_p - X_p \tag{7}$$

Adaptation laws:

$$\dot{\hat{A}}_p = -\gamma_1 P e X_p^T \quad , \quad \dot{\hat{B}}_p = -\gamma_2 P e u^T \tag{8}$$

The matrix P is obtained using the selected A_m and Q from:

$$P A_m + A_m^T P = -Q = -I$$

which results in:

$$P = \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}$$

To determine the values of γ_1 and γ_2, we increase them in a way that minimizes the system's settling time.

Also, since there are 6 parameters to estimate, using 3 sinusoidal inputs will be sufficient.

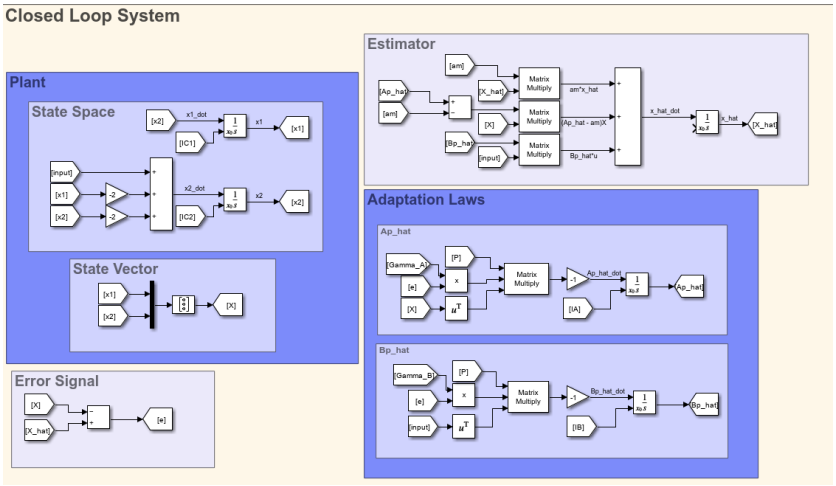


Figure 4: Overall closed-loop structure for Question 2 in Simulink

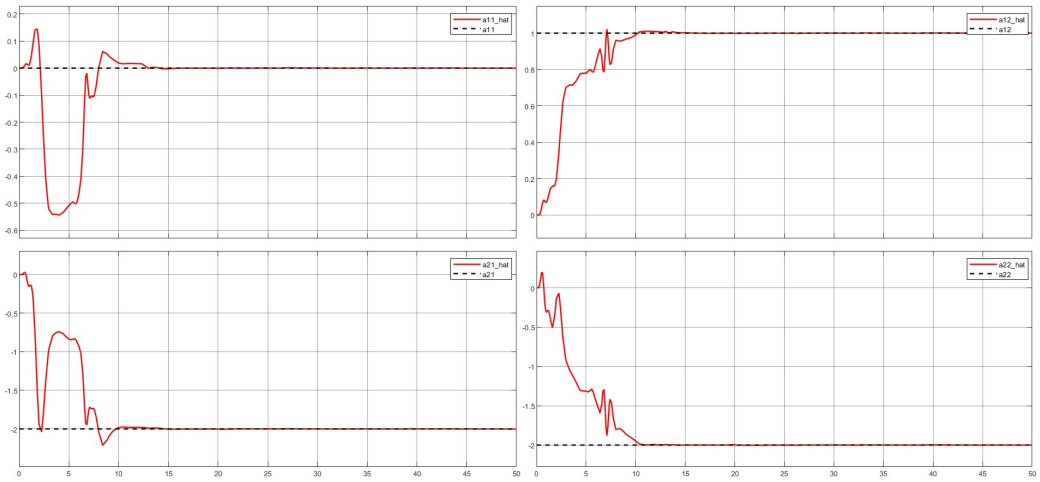


Figure 5: Parameter estimation $\hat{A}_p$

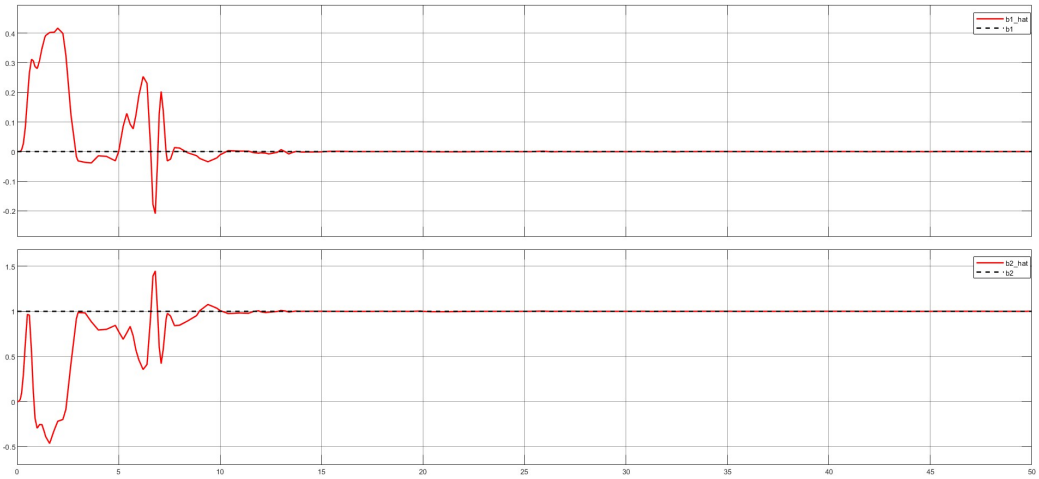


Figure 6: Parameter estimation $\hat{B}_p$